Microeconomics

1. Suppose the demand function for good X is given by:

$$Q_x = 200 - 4P_x + 2P_y + 0.05I$$

where P_x is the price of good X, P_y is the price of a related good, and *I* is consumer income.

Assume that P_x =20, P_y =30 , and *I* =800.

a) Calculate the quantity demanded of good X.

b) Using differentiation, calculate the price elasticity of demand for good X.

c) Calculate the income elasticity of demand for good X.

d) Calculate the cross-price elasticity of demand for good X with respect to the price of good Y

e) Interpret your results. Is good X a normal or inferior good? Are goods X and Y substitutes or complements?

2. A firm's long-run total cost function is given by:

$$TC(Q) = 500 + 20Q + 2Q^2$$

where TC represents total cost and Q represents the quantity of output produced.

Questions

a) Derive the firm's average cost (AC) and marginal cost (MC) functions.

b) Using differentiation, determine the range of output over which the firm experiences economies of scale.

c) Determine the output level at which average cost is minimized.

d) Explain the relationship between economies of scale, constant economies of scale, and diseconomies of scale.

3. Consider the following production functions, where Q represents total output, and X and Y represent inputs used in production.

(a) $Q = 4X + 6Y$
(b) $Q = 5X^{0.4}Y^{0.8}$
(c) $Q = 12X^{0.2}Y^{0.5}$

For each production function, determine whether it exhibits increasing, constant, or decreasing returns to scale.

4. Given:

$$U = Q_1 Q_2$$

a) Maximize utility when P_1= 1, P_2= 4, and one's budget B=120

b) Estimate the effect of a 1-unit increase in the budget.

5. A firm produces output using two inputs: capital (K) and labor (L). The firm's production function is given by:

$$Q = 8K^{0.25}L^{0.75}$$

The rental rate of capital is $20 per unit, and the wage rate of labor is $10 per unit. The firm has a total production budget of $800.

a) Formulate the firm's output maximization problem subject to its total cost constraint.

b) Set up the Lagrangian function and derive the first-order conditions using partial differentiation.

c) Determine the optimal quantities of capital and labor that maximize output.

d) Calculate the maximum level of output the firm can produce.